

INTRODUCTION TO C

C IS A

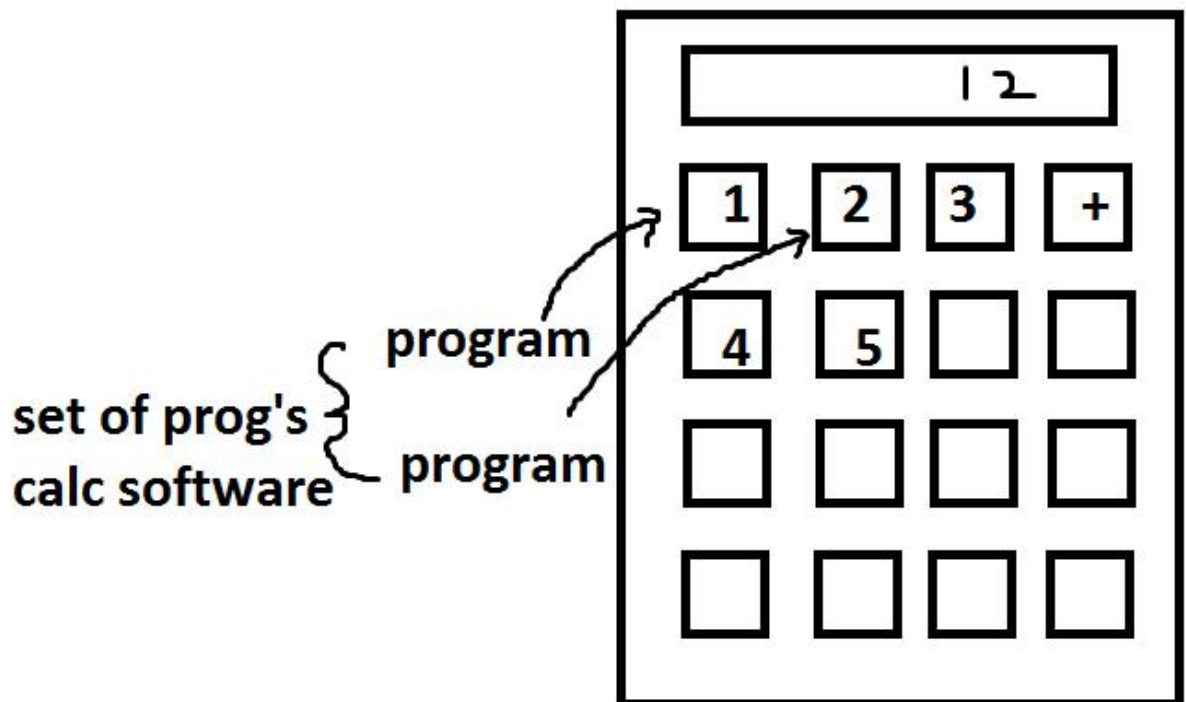
1. It is a high level / middle level programming language.
2. C is a Compiler based programming language.

What is a program?

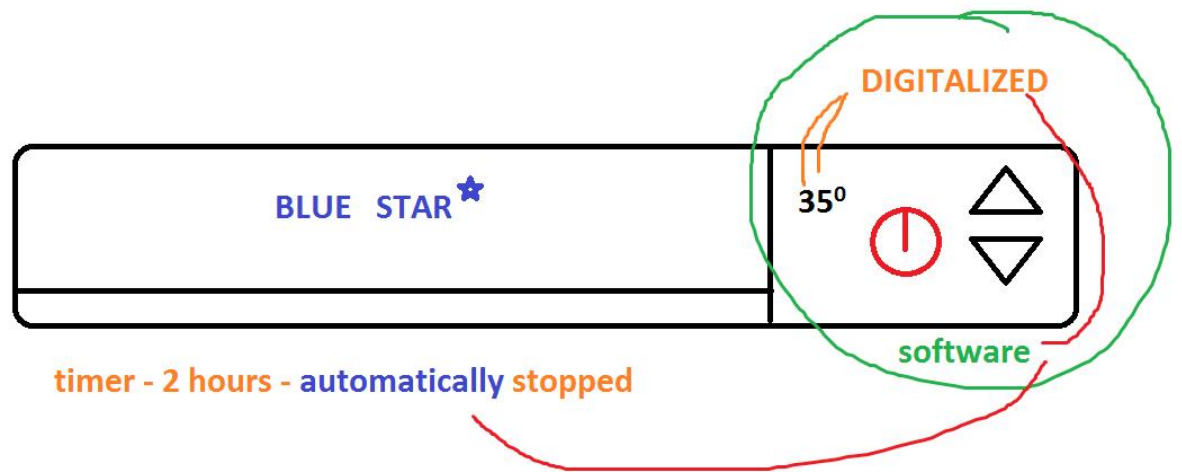
Set of instructions is called program

What is a software?

Set of programs is called software.



As per IT Industry software is a digitalized and automated process.



Basically the software divided into 2 types.

1. System software

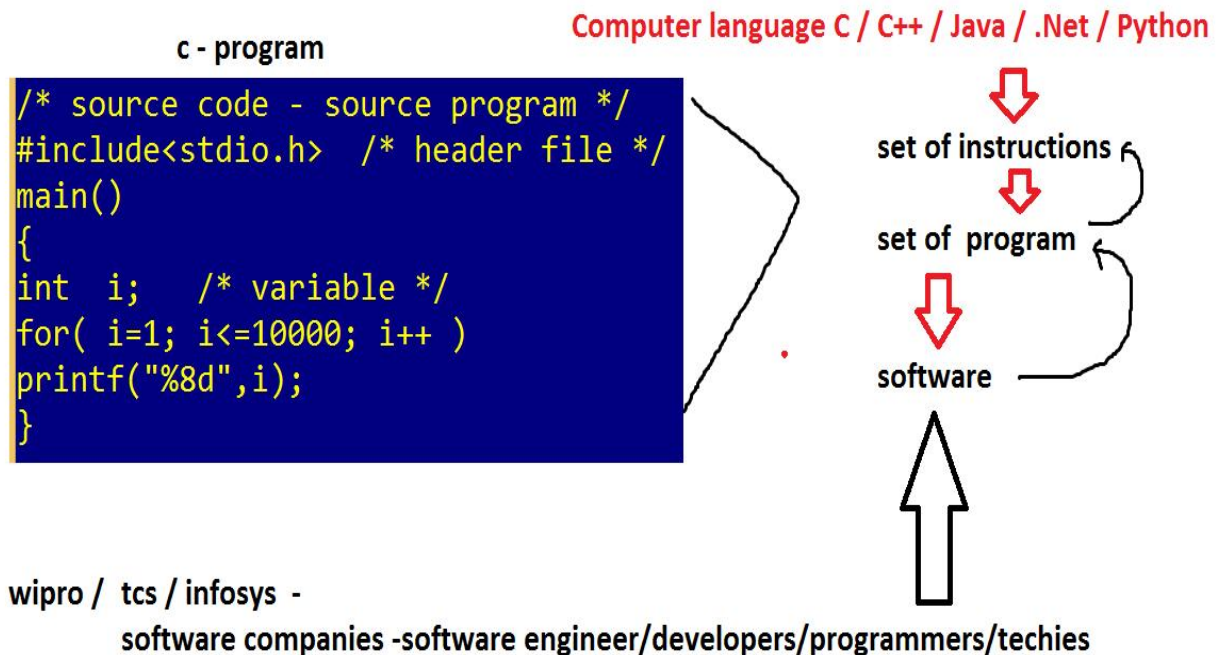
Eg: operating system, device drivers, translators

2. Application software

Eg: whatsapp, phonepe, irctc,...

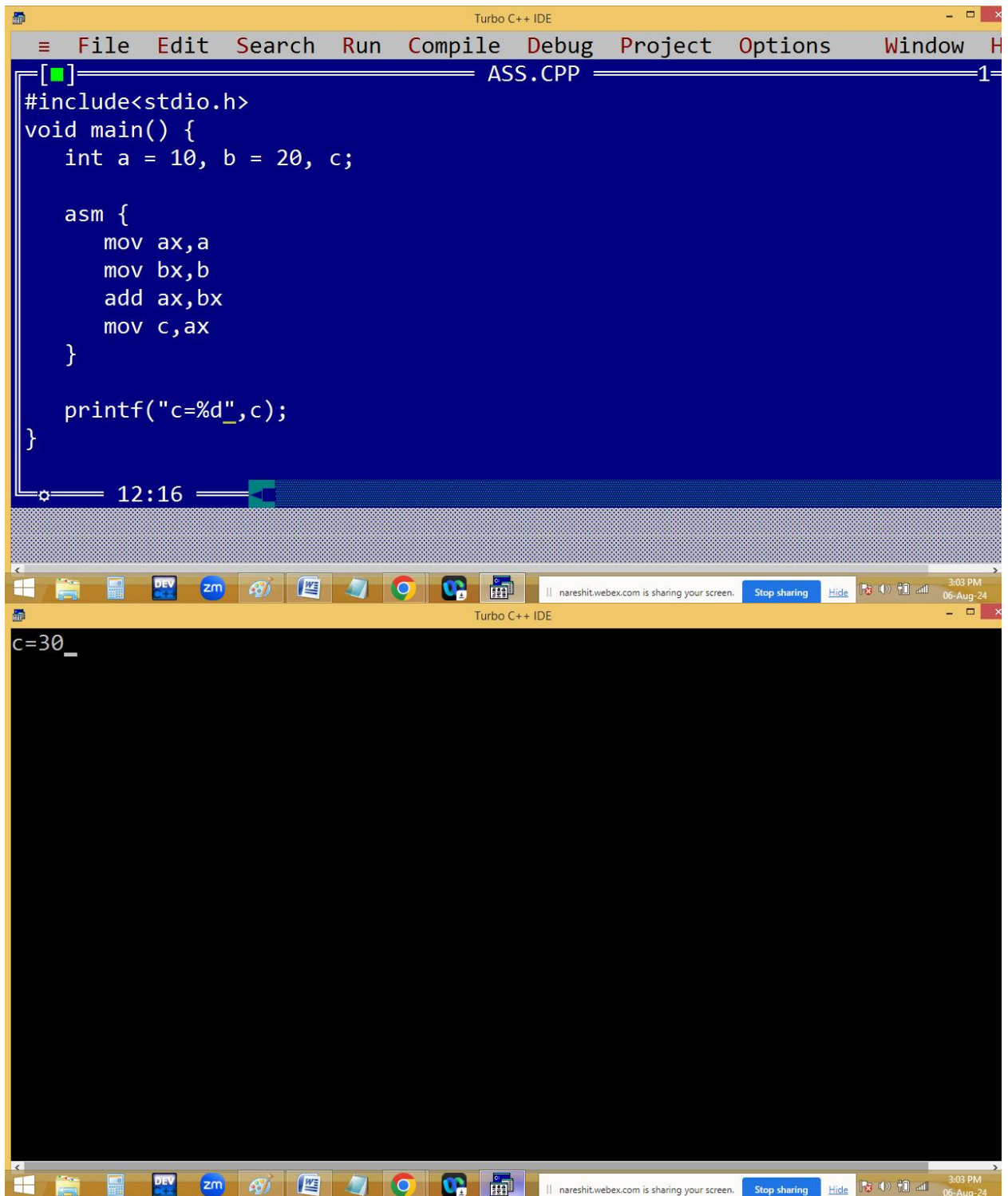
What is a language? Generally the languages like telugu / English / hindi / Marathi etc are human languages and they are used to communicate with the humans. To communicate with the computers we have to use the **computer programming languages** like C / C++ / Java / .Net / Python etc. By using all these languages we are

creating the software [programs] to **communicate with the machines.**



Basically the computer languages are divided into 3 types.

1. **Machine language**: Created with binary code[0, 1]
Eg: 10111100
2. **Low level / assembly language**: Created with English like shortcuts called **MNEMONICS**.
Eg: gd mrg, sub, add, ..



The image shows a screenshot of the Turbo C++ IDE. The top window, titled 'ASS.CPP', contains the following C code:

```
[■] ASS.CPP 1
#include<stdio.h>
void main() {
    int a = 10, b = 20, c;

    asm {
        mov ax,a
        mov bx,b
        add ax,bx
        mov c,ax
    }

    printf("c=%d_",c);
}
```

The bottom window shows the output of the program: `c=30_`. The Windows taskbar at the bottom includes icons for various applications and a system tray showing the time as 3:03 PM on 06-Aug-24.

3. High level language: Created with simple english and easy to understand.

Eg: good morning, subject, subtract, subway,
subordinate, subscribe, substitute, addition, address,
addiction, ...

Turbo C++ IDE

File Edit Search Run Compile Debug Project Options Window Help

NONAME00.CPP 2

```
[■]
#include<stdio.h>
void main()
{
int a=10, b=20, c=a+b;
printf("Sum=%d",c);
}
```

Turbo C++ IDE

Sum=30_

C comes with both low level and high level features. Hence it is also called middle level language.

C **low level features** used to develop **system software**.

High level features used to develop **application software**.

Hence c is a **multi-purpose** programming language.

What is a translator?

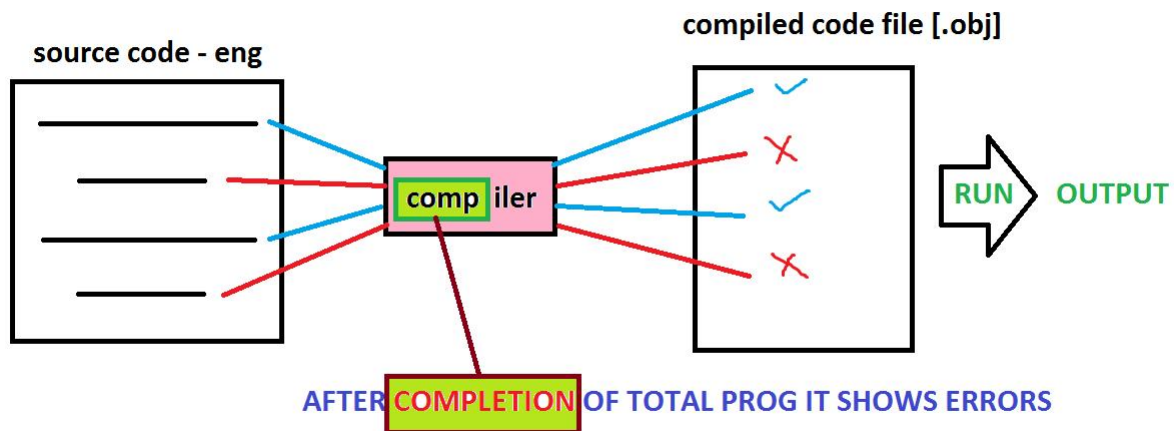


Always the user given instructions are in English, which is called source code / source program. But the computer understandable code is binary code. **To convert this source code into binary code and to check the errors [programming mistakes] we are using the translators like**

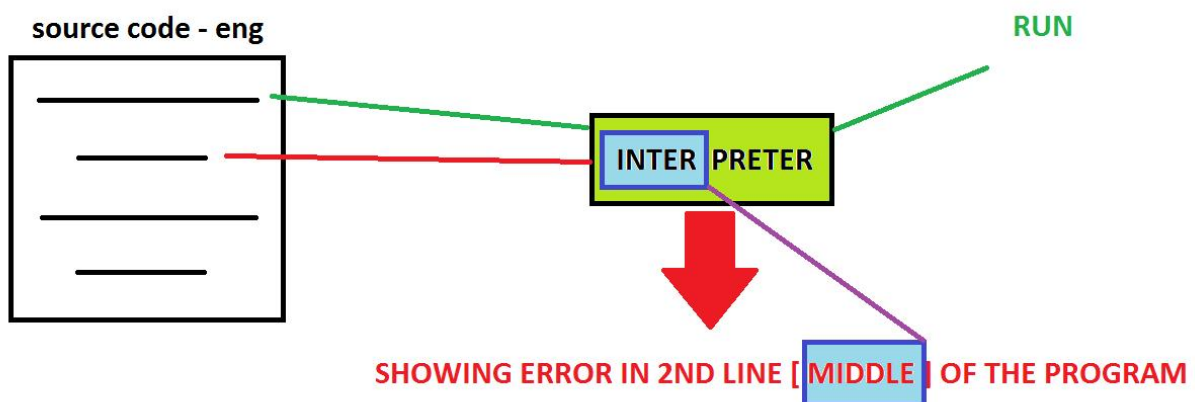
1. Compiler
2. Interpreter
3. Assembler

Compiler and interpreter both used to convert **high level** programs to **machine language**.

Compiler converts the total source code into binary code **at once** by leaving error lines.



Interpreter converts **line by line**.



Assembler used to convert **low level** programs to machine language and working style is similar to compiler.